WORKING DRAFT / DEMO TODAY

Resin 3D Printing Artist Guide

Oolite Arts Digital Lab - Anycubic Photon Mono M7 Max launch packet

Purpose: a lean, artist-facing guide for preparing first resin prints before the new Digital Lab space fully launches. This is not the full installation manual; it focuses on what artists need to know, what staff will verify, and what can cause delays or failed prints.

PHOTO NEEDED

Hero photo: printer + clean workstation

Meeting demo framing

- Materials are in hand and first prints are being tested.
- Launch phase should remain staff-assisted until the workflow is stable.
- Artists should submit printable files and understand safety, size, support, and timing limits.
- Video QR references setup chapters for staff and context.

Suggested public language: The Digital Lab is beginning staff-assisted resin 3D print testing. Resident artists may prepare files for review; final print approval depends on file readiness, material availability, safe operation, and staff capacity.

QR placeholder: video reference for staff chapters / setup context

1. Artist-facing workflow

1 Request → 2 File check → 3 Slice → 4 Print → 5 Wash + cure → 6 Pickup

Staff-assisted resin printing workflow for launch period. Artists prepare files; Digital Lab confirms printability and operates machine.

Artist prepares

- 3D model file: STL / OBJ / supported slicer file.
- Desired size in inches or millimeters.
- Material preference if available: standard gray, clear, tough/ABS-like, etc.
- Deadline or event context, clearly stated.

Digital Lab verifies

- Print fits build volume.
- Model is watertight, scaled correctly, and orientable.
- Supports, hollowing, drain holes, wall thickness, and resin use are reasonable.
- Estimated print + wash + cure time.

Launch policy recommendation: During the first-print phase, artists do not self-operate the resin printer. Staff handles resin, slicing confirmation, machine setup, post-processing, and cleaning until a formal training / authorization workflow is approved.

Images needed on this page

WORKFLOW PHOTO

Artist file review at computer

STAFF OPERATION PHOTO

Printer in staff-only mode

2. Before you submit a file

A fast resin print starts with a clean file and clear intent. These checks prevent wasted resin, failed prints, and avoidable scheduling issues.

File question	Artist answer needed
What is the final size?	Width x depth x height, with units.
Is it solid, hollow, or shell-like?	If hollow, drain holes are likely required.
Does it need supports?	Most sculptural / overhanging forms need supports. Staff will confirm.
What surface matters most?	Tell staff which side should look cleanest. Support marks may appear.
Is it fragile or thin?	Thin details may fail during print, wash, support removal, or cure.
Is it urgent?	State deadline. Resin printing is not same-day by default.

Artist tip: Resin printing rewards planning. The machine can produce fine detail, but scale, supports, wall thickness, resin choice, washing, and curing all affect the final object.

Image set for this page

GOOD FILE

watertight + scaled

RISKY FILE

thin + unsupported

3. What this printer is good for

Strong fits

- Small sculptures and maquettes with fine details.
- Masks, relief objects, props, figurative or ornamental elements.
- High-detail prototypes that need smoother surfaces than FDM.
- Transparent, gray, tough, or specialty resin tests when approved.

Not ideal / needs review

- Food-safe objects or objects for skin contact unless material safety is confirmed.
- Large solid blocks that waste resin, get heavy, or cure poorly.
- Very thin wires, unsupported fingers, delicate lattices, or brittle details.
- Objects that must be structurally load-bearing without material testing.

Machine reference - for staff and artist expectations

Build volume	298 x 164 x 300 mm
LCD / resolution context	13.6 inch 7K LCD; approx. 46 x 46 micron XY resolution
Resin vat	1.3 L vat with scale lines; auto-fill and retrieval system
Heating	Resin temperature control range noted in video: 20-40 C
Leveling	Manual four-point leveling, handled by staff

Source note: The video identifies this printer as the Anycubic Photon Mono M7 Max and gives the build volume, resin vat, heating, and manual leveling details. Official Anycubic product materials also list the 298 x 164 x 300 mm print size.

4. Safety and space rules artists need to know

Resin printing is different from filament printing. The uncured resin, IPA cleaning process, UV exposure, odor, and contaminated tools require controlled handling.

Required baseline

- Gloves when near uncured resin, wet prints, vat, build plate, or IPA station.
- Eye protection during UV exposure tests and resin handling.
- Mask / ventilation plan for odor and fume control.
- Dedicated contaminated tools: scrapers, trays, towels, gloves, and disposal container.

Artist boundaries

- Do not pour resin unless staff authorizes and supervises.
- Do not open the printer mid-print.
- Do not touch uncured prints, resin vat, build plate, or LCD film.
- Do not remove supports until the staff-approved stage.

Gotcha from the video: odor and air quality can become noticeable during cleaning and resin handling. The video also shows that resin can splash during autofill testing, so leak checks and feed tests should be staff-only and done with protection.

PPE PHOTO

gloves / glasses / mask

VENTILATION PHOTO

printer + purifier / airflow

5. Print day: what happens after approval

Staff print-day sequence

- Confirm resin bottle, vat, build plate, and machine status.
- Run self-check / heating / feed settings as needed.
- Start print from USB, cloud, or approved slicer workflow.
- Remove finished build plate and drain safely.
- Wash object, cure object, clean build plate, clean station.

What artists should expect

- Prints may need support removal and light finishing after curing.
- Support marks are normal and may need sanding or painting.
- Hollow objects need drainage and may leak if not washed/cured correctly.
- Pickup timing depends on print duration plus wash/cure/cleaning time.

Post-processing stages

Stage	What happens	Artist note
Drain	Wet object drips resin back into vat or tray.	Do not touch.
Wash	Object is rinsed/washed to remove uncured resin.	Hollow pieces must drain.
Cure	UV cure hardens the object.	Over-curing can make some resins brittle.
Finish	Supports, sanding, priming, painting if needed.	Finish quality depends on design + supports.

6. Common gotchas that delay first prints

Scale mismatch	File arrives in the wrong units; object becomes tiny or oversized.
Unsupported overhangs	Beautiful on screen, but fails without supports or better orientation.
No drain holes	Hollow object traps liquid resin and becomes a safety/problem object.
Too solid / too heavy	Uses too much resin, increases failure risk, and may cure poorly.
Thin details	Hands, hair, antennae, text, spikes, and mesh details may break.
Cloud / Wi-Fi issues	The video shows cloud binding/network reset can be annoying; USB fallback should be ready.
Cleaning station size	Large build plate / large object may not fit standard rinse containers; plan post-processing before printing.

Staff note for first launch

For the first group of artists, prioritize successful small-to-medium demo objects over maximum-size prints. This creates clean proof of workflow, avoids unnecessary resin waste, and gives the Digital Lab documentation photos for the final public guide.

FAILED PRINT EXAMPLE

show risk without blame

SUCCESSFUL TEST PRINT

clean object + notes

7. Visual guide production plan

The final artist packet should be highly visual and modular. Capture these images during the first-print workflow so the guide becomes Oolite-specific instead of generic.

Hero / orientation	Printer in the Digital Lab, clean desk, PPE visible, no clutter.
File readiness	Computer screen with model preview, scale callout, support/orientation diagram.
Safety setup	Gloves, glasses, mask, IPA/wash area, resin bottle, covered waste zone.
Machine boundaries	Artist side vs staff-only resin handling area.
Print lifecycle	Before / during / after: build plate, washed object, cured object.
Gotchas	Thin detail, support marks, hollow/drain holes, scale comparison object.
QR references	Video chapters, official manual, booking/request form, Digital Lab email.

Video chapter map to connect with QR references

0:31	Printer capacity and scale
6:41	Initial setup / leveling
13:10	UV exposure test
16:40	Configuration / USB / cloud
19:02	Heating / feed / retrieval
22:14	Print removal
25:03	Cleaning
30:44	Troubleshooting network

Replace / confirm QR link before public release.

Add final booking form QR once approved.

Contact: digilab@oolitearts.org

8. One-page artist checklist

Use this as the front desk / form checklist for first print requests.

■	File submitted in STL / OBJ / supported format.
■	Final object size stated with units.
■	Artist confirms which surface should be cleanest.
■	Artist understands supports may leave marks.
■	Staff reviewed model orientation, supports, hollowing, and drain holes.
■	Material / resin is available and appropriate.
■	Print time + wash/cure + pickup window estimated.
■	PPE, ventilation, cleaning, and waste workflow ready.
■	USB fallback prepared in case app/cloud connection fails.
■	Final approval documented before print begins.

Draft form fields to add later

Artist name / email / studio; file link; object dimensions; resin preference; deadline; staff review status; approval date; print started; washed; cured; picked up.

Sources: Video transcript supplied by Moises Sanabria for the Anycubic Photon Mono M7 Max setup/demo; official Anycubic product/wiki materials checked for printer specs and safety context. Final public release should verify all current equipment, resin, PPE, ventilation, and booking policies with Oolite leadership.